

Next Article:

[Difference Between Vector and List](#) →

Difference between ELT and ETL

Last Updated : 18 Mar, 2025



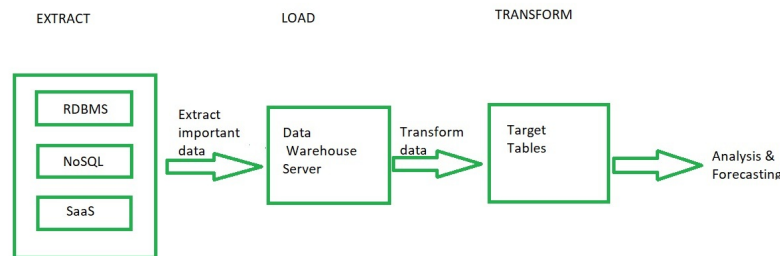
In managing and analyzing data, two primary approaches i.e. ETL (Extract, Transform, Load) and ELT (Extract, Load, Transform), are commonly used to move data from various sources into a data warehouse. Understanding the differences between these methods is crucial for selecting the right approach based on our data needs, storage system and performance requirements.

ELT Process

Extraction, Load and Transform (ELT) is the technique of extracting raw data from the source, storing it in the [data warehouse](#) of the target server and preparing it for end-stream users.

ELT consists of three different operations performed on the data:

1. **Extract:** Extracting data is the process of identifying data from one or more sources. The sources may include databases, files, [ERP](#), [CRM](#), or any other useful source of data.
2. **Load:** Loading is the process of storing the extracted raw data in a data warehouse or [data lake](#).
3. **Transform:** [Data transformation](#) is the process in which the raw data from the source is transformed into the target format required for analysis

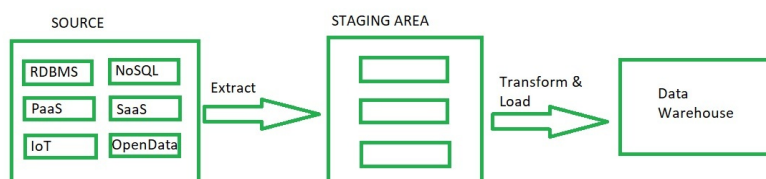


Data from the sources is extracted and stored in the data warehouse. The entire data is not transformed; only the required transformations are done when necessary. Raw data can be retrieved from the warehouse anytime when required. The data, transformed as needed, is then sent forward for analysis. When we use ELT, we move the entire data set as it exists in the source systems to the target. This means that we have the raw data at your disposal in the data warehouse, in contrast to the ETL approach.

ETL Process

[ETL](#) is the traditional technique of extracting raw data, transforming it as required for the users and storing it in data warehouses. ELT was later developed, with ETL as its base. The three operations in ETL and ELT are the same, except that their order of processing is slightly different. This change in sequence was made to overcome some drawbacks.

1. **Extract:** It is the process of extracting raw data from all available data sources such as databases, files, ERP, CRM or any other.
2. **Transform:** The extracted data is immediately transformed as required by the user.
3. **Load:** The transformed data is then loaded into the data warehouse from where the users can access it.



The data collected from the sources is directly stored in the staging area. The required transformations are performed on the data in the staging area. Once the data is



the [ETL architecture](#) is that once the transformed data is stored in the warehouse, it cannot be modified again. In contrast, in ELT, a copy of the raw data is always available in the warehouse and only the required data is transformed when needed.

Difference between ELT and ETL

Category	ETL	ELT
Acronym Meaning	Extract, Transform, Load	Extract, Load, Transform
Definition	Extracts raw data, transforms it on a secondary server, then loads it into the destination.	Extracts raw data, loads it directly into the destination and transforms it there.
Processing Speed	Slower; data transformation occurs before loading.	Faster; data is loaded first and transformed in parallel.
Data Volume	Best for smaller, complex data sets like marketing data.	Suited for large data sets requiring speed, like real-time analytics.
Data Output	Primarily structured data.	Structured, semi-structured and unstructured data.
Data Lake Compatibility	Not compatible with data lakes.	Fully compatible with data lakes.
Maturity	Well-established, used for 20+ years, with extensive documentation.	Newer approach with fewer tools and less documentation.
Cost Efficiency	Higher costs due to the need for separate servers and processing infrastructure.	More cost-effective, leveraging cloud resources for scalability.
Security	Requires custom security solutions to protect sensitive data.	Built-in security features like access control and multifactor authentication.
Transformation Location	Data is transformed on a secondary server before loading.	Data is loaded as-is and transformed within the target system.
Flexibility	Best for structured data transformation.	Handles structured and unstructured data with ease.

Similarities Between ETL and ELT

Both ETL (Extract, Transform, Load) and ELT (Extract, Load, Transform) are data integration processes that consolidate data from various sources into a single, unified repository for further analysis. They share several key similarities:

- **Data Extraction:** Both processes begin by extracting raw data from multiple sources like databases, files, [SaaS applications](#), or [IoT devices](#). This data can be structured, semi-structured, or unstructured.
- **Data Transformation:** While the timing of transformation differs, both ETL and ELT involve transforming the extracted data into a format that aligns with the target system's requirements. This ensures data is clean, accurate and ready for analysis.
- **Data Loading:** Both methods ultimately load the processed data into a data

- **Unified Data Repository:** Both processes help create a single source of truth, ensuring that enterprise data is consistent, accurate and up-to-date for decision-making.

Choosing Between ETL and ELT

The choice between ETL and ELT depends on our specific needs and requirements.

- ETL works well for smaller datasets and [structured data](#) where the data needs to be transformed immediately. It often requires special hardware and can be less flexible when handling large amounts of data.
- ELT is better for large datasets and unstructured or [non-relational data](#). It is more flexible and cost-effective, especially with cloud-based data solutions. With ELT, we can store raw data and transform it as needed.

Difference between ELT and ETL – FAQs

What's the main difference between ETL and ELT?

ETL transforms the data before loading it into the warehouse, while ELT loads the raw data first and transforms it later. ELT is more flexible and can handle larger datasets better.

Can ELT handle structured data?

Yes, ELT can handle both structured and unstructured data. It's especially useful for large volumes of data, whether it's structured or not.

What types of storage systems are used in ELT?

ELT often uses modern storage systems like Hadoop or NoSQL databases, but it can also work with traditional RDBMS systems.

Are there any disadvantages to using ELT?

While ELT is flexible, it may require more storage space because it keeps raw data. It also might need more advanced data management skills to handle transformations effectively.

 Comment

More info 

Advertise with us

Next Article 

Difference Between Vector and List

Similar Reads

Difference between ELT and ETL

In managing and analyzing data, two primary approaches i.e. ETL (Extract, Transform, Load) and ELT (Extract, Load, Transform), are commonly used to move data from various sources into a data warehouse. Understanding the...

 6 min read

Difference Between Vector and List

Vector: Vector is a type of dynamic array which has the ability to resize automatically after insertion or deletion of elements. The elements in vector are placed in contiguous storage so that they can be accessed and traversed...

 2 min read

Difference between JFET and MOSFET

🕒 2 min read

Difference Between EDG And EWG

The EDG and EWG, both are electrophilic aromatic directing groups. And these are forms of substituents that we can find in organic compounds. If you've looked at a molecule's structure, you might have noticed that some atom...

🕒 4 min read

Difference between ERP and EAM

Enterprise Resource Planning (ERP) and Enterprise Asset Management (EAM) are important structures utilized by organizations to streamline operations and manage resources efficiently. While ERP specializes in the overall...

🕒 4 min read

Difference between ES5 and ES6

ECMAScript (ES) is a standardized scripting language specification developed by Ecma International. It serves as the foundation for JavaScript and many other scripting languages. Over time, different versions of ECMAScript ha...

🕒 3 min read

Difference between IoE and IoT

1. Internet of Everything (IoE) : IoE is the intelligent connection between 4 key elements i.e people, process, data, and things. It is considered as superset of Internet of Things (IoT). IoE covers the wider concept of connectivity...

🕒 4 min read

Difference between LTE and VoLTE

In the case of mobile networks, you may have come across acronyms like LTE or VoLTE at some point. Both are associated with the connectivity of your phone to the network and the way you receive and make calls, yet they...

🕒 5 min read

Difference between LED and LCD

Light Emitting Diode (LED): LED is a type of LCD that actually accompanies the advancement of technology. This replaces the fluorescent tube with backlight technology, which produces a clearer picture than the LCD. LED have...

🕒 2 min read

Difference between LXC and LXD

LXC stands for Linux Containers, it is a tool that is used for the virtualization of operating systems. By using LXC one can operate any software or application in a virtual environment rather than on the physical system. The virtu...

🕒 2 min read



📍 Corporate & Communications Address:
A-143, 7th Floor, Sovereign Corporate
Tower, Sector- 136, Noida, Uttar Pradesh
(201305)

📍 Registered Address:
K 061, Tower K, Gulshan Vivante
Apartment, Sector 137, Noida, Gautam
Buddh Nagar, Uttar Pradesh, 201305



Advertise with us

Company

About Us
Legal
Privacy Policy
In Media
Contact Us
Advertise with us
GFG Corporate Solution
Placement Training
Program
GeeksforGeeks
Community

Computer Science

Operating Systems
Computer Network
Database Management
System
Software Engineering
Digital Logic Design
Engineering Maths
Software Development
Software Testing

Languages

Python
Java
C++
PHP
GoLang
SQL
R Language
Android Tutorial
Tutorials Archive

DevOps

Git
Linux
AWS
Docker
Kubernetes
Azure
GCP
DevOps Roadmap

DSA

Data Structures
Algorithms
DSA for Beginners
Basic DSA Problems
DSA Roadmap
Top 100 DSA Interview
Problems
DSA Roadmap by
Sandeep Jain
All Cheat Sheets

System Design

High Level Design
Low Level Design
UML Diagrams
Interview Guide
Design Patterns
OOAD
System Design
Bootcamp
Interview Questions

Data Science & ML

Data Science With
Python
Data Science For
Beginner
Machine Learning
ML Maths
Data Visualisation
Pandas
NumPy
NLP
Deep Learning

Interview Preparation

Competitive
Programming
Top DS or Algo for CP
Company-Wise
Recruitment Process
Company-Wise
Preparation
Aptitude Preparation
Puzzles

Web Technologies

HTML
CSS
JavaScript
TypeScript
ReactJS
NextJS
Bootstrap
Web Design

School Subjects

Mathematics
Physics
Chemistry
Biology
Social Science
English Grammar
Commerce
World GK

Python Tutorial

Python Programming
Examples
Python Projects
Python Tkinter
Web Scraping
OpenCV Tutorial
Python Interview
Question
Django

GeeksforGeeks

Videos

DSA
Python
Java
C++
Web Development
Data Science
CS Subjects

@GeeksforGeeks, Sanchhaya Education Private Limited, All rights reserved